UNSUPERVISED LEARNING

Hierarchical Clustering

10 data points · 9 merges · a new distance matrix after every merge

SINGLE LINKAGE

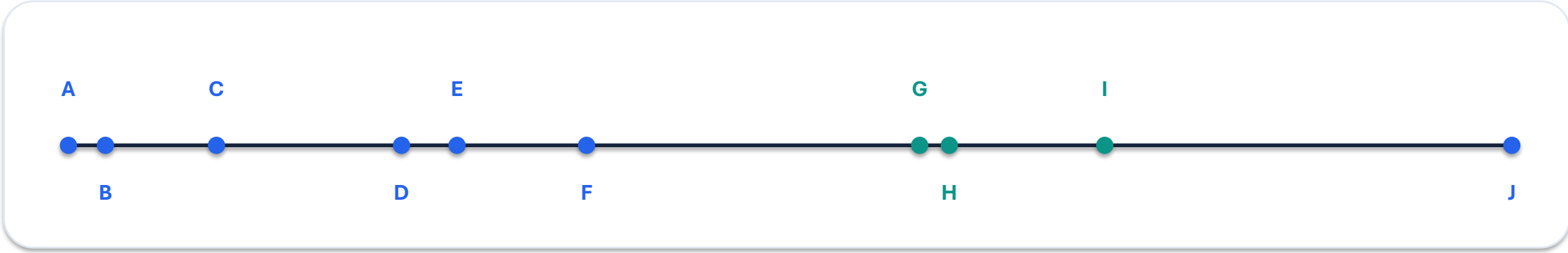
point + point → cluster + point → cluster + cluster

10 points on one number line

We use one feature so every distance can be calculated by hand.

A	B	C	D	E	F	G	H	I	J
1	2	5	10	11.5	15	24	24.8	29	40

Values are ordered from left to right:



Starting clusters: A | B | C | D | E | F | G | H | I | J

Single Linkage: always use the nearest cross-cluster pair

After each merge, recalculate the distance matrix using the new clusters.

POINT ↔ POINT

$$d(A,B) = |1 - 2| = 1$$

Absolute difference of two values.

CLUSTER ↔ POINT

$$d(AB,C) = \min(4, 3) = 3$$

Point C joins cluster AB.

CLUSTER ↔ CLUSTER

$$d(ABC,DEF) = 5$$

Closest cross-cluster pair is C–D.

Merge G + H at distance 0.8

Current distance matrix: 10 clusters → 9 clusters

CURRENT DISTANCE MATRIX

–	A	B	C	D	E	F	G	H	I	J
A	0	1	4	9	10.5	14	23	23.8	28	39
B	1	0	3	8	9.5	13	22	22.8	27	38
C	4	3	0	5	6.5	10	19	19.8	24	35
D	9	8	5	0	1.5	5	14	14.8	19	30
E	10.5	9.5	6.5	1.5	0	3.5	12.5	13.3	17.5	28.5
F	14	13	10	5	3.5	0	9	9.8	14	25
G	23	22	19	14	12.5	9	0	**0.8**	5	16
H	23.8	22.8	19.8	14.8	13.3	9.8	**0.8**	0	4.2	15.2
I	28	27	24	19	17.5	14	5	4.2	0	11
J	39	38	35	30	28.5	25	16	15.2	11	0

Teal cells mark the smallest non-zero distance.

TWO INDIVIDUAL POINTS

DISTANCE CALCULATION

$$d(G,H) = |24 - 24.8| = 0.8$$

RESULTING CLUSTERS

A, B, C, D, E, F, GH, I, J

Next: d(GH,I) = min(5, 4.2) = 4.2

Merge A + B at distance 1

Current distance matrix: 9 clusters → 8 clusters

CURRENT DISTANCE MATRIX

–	A	B	C	D	E	F	GH	I	J
A	0	1	4	9	10.5	14	23	28	39
B	1	0	3	8	9.5	13	22	27	38
C	4	3	0	5	6.5	10	19	24	35
D	9	8	5	0	1.5	5	14	19	30
E	10.5	9.5	6.5	1.5	0	3.5	12.5	17.5	28.5
F	14	13	10	5	3.5	0	9	14	25
GH	23	22	19	14	12.5	9	0	4.2	15.2
I	28	27	24	19	17.5	14	4.2	0	11
J	39	38	35	30	28.5	25	15.2	11	0

Teal cells mark the smallest non-zero distance.

TWO INDIVIDUAL POINTS

DISTANCE CALCULATION

$$d(A,B) = |1 - 2| = 1$$

RESULTING CLUSTERS

AB, C, D, E, F, GH, I, J

Next: d(AB,C) = min(4, 3) = 3

Merge D + E at distance 1.5

Current distance matrix: 8 clusters → 7 clusters

CURRENT DISTANCE MATRIX

–	AB	C	D	E	F	GH	I	J
AB	0	3	8	9.5	13	22	27	38
C	3	0	5	6.5	10	19	24	35
D	8	5	0	**1.5**	5	14	19	30
E	9.5	6.5	**1.5**	0	3.5	12.5	17.5	28.5
F	13	10	5	3.5	0	9	14	25
GH	22	19	14	12.5	9	0	4.2	15.2
I	27	24	19	17.5	14	4.2	0	11
J	38	35	30	28.5	25	15.2	11	0

Teal cells mark the smallest non-zero distance.

TWO INDIVIDUAL POINTS

DISTANCE CALCULATION

$$d(D,E) = |10 - 11.5| = 1.5$$

RESULTING CLUSTERS

AB, C, DE, F, GH, I, J

Next: d(DE,F) = min(5, 3.5) = 3.5

Merge AB + C at distance 3

Current distance matrix: 7 clusters → 6 clusters

CURRENT DISTANCE MATRIX

–	AB	C	DE	F	GH	I	J
AB	0	3	8	13	22	27	38
C	3	0	5	10	19	24	35
DE	8	5	0	3.5	12.5	17.5	28.5
F	13	10	3.5	0	9	14	25
GH	22	19	12.5	9	0	4.2	15.2
I	27	24	17.5	14	4.2	0	11
J	38	35	28.5	25	15.2	11	0

Teal cells mark the smallest non-zero distance.

POINT JOINS A CLUSTER

DISTANCE CALCULATION

$$d(AB,C) = \min(d(A,C), d(B,C)) = \min(4, 3) = 3$$

RESULTING CLUSTERS

ABC, DE, F, GH, I, J

New cluster: ABC = A + B + C

Merge DE + F at distance 3.5

Current distance matrix: 6 clusters → 5 clusters

CURRENT DISTANCE MATRIX

–	ABC	DE	F	GH	I	J
ABC	0	5	10	19	24	35
DE	5	0	**3.5**	12.5	17.5	28.5
F	10	**3.5**	0	9	14	25
GH	19	12.5	9	0	4.2	15.2
I	24	17.5	14	4.2	0	11
J	35	28.5	25	15.2	11	0

Teal cells mark the smallest non-zero distance.

POINT JOINS A CLUSTER

DISTANCE CALCULATION

$$d(DE,F) = \min(d(D,F), d(E,F)) = \min(5, 3.5) = 3.5$$

RESULTING CLUSTERS

ABC, DEF, GH, I, J

New cluster: DEF = D + E + F

Merge GH + I at distance 4.2

Current distance matrix: 5 clusters → 4 clusters

CURRENT DISTANCE MATRIX

–	ABC	DEF	GH	I	J
ABC	0	5	19	24	35
DEF	5	0	9	14	25
GH	19	9	0	**4.2**	15.2
I	24	14	**4.2**	0	11
J	35	25	15.2	11	0

Teal cells mark the smallest non-zero distance.

POINT JOINS A CLUSTER

DISTANCE CALCULATION

d(GH,I) = min(d(G,I), d(H,I))
= min(5, 4.2) = 4.2

RESULTING CLUSTERS

ABC, DEF, GHI, J

New cluster: GHI = G + H + I

A cluster can merge with another cluster

Before merge 7: compare every point in ABC with every point in DEF.

DISTANCES BETWEEN ABC AND DEF

	D (10)	E (11.5)	F (15)
A (1)	9	10.5	14
B (2)	8	9.5	13
C (5)	5	6.5	10

SINGLE LINKAGE

Select the smallest of all nine distances.

Minimum = 5

C (5) and D (10)
are the closest pair.

Merge ABC and DEF next.

Merge ABC + DEF at distance 5

Current distance matrix: 4 clusters → 3 clusters

CURRENT DISTANCE MATRIX

–	ABC	DEF	GHI	J
ABC	0	5	19	35
DEF	5	0	9	25
GHI	19	9	0	11
J	35	25	11	0

Teal cells mark the smallest non-zero distance.

TWO CLUSTERS MERGE

DISTANCE CALCULATION

d(ABC,DEF) = 5
Closest pair: C and D
|5 – 10| = 5

RESULTING CLUSTERS

ABCDEF, GHI, J

Check all 3 × 3 cross-cluster pairs

Merge ABCDEF + GHI at distance 9

Current distance matrix: 3 clusters → 2 clusters

CURRENT DISTANCE MATRIX

–	ABCDEF	GHI	J
ABCDEF	0	**9**	25
GHI	**9**	0	11
J	25	11	0

Teal cells mark the smallest non-zero distance.

TWO CLUSTERS MERGE

DISTANCE CALCULATION

d(ABCDEF,GHI) = 9
Closest pair: F and G
|15 − 24| = 9

RESULTING CLUSTERS

ABCDEFGHI, J

Single Linkage uses the closest cross-pair

Merge ABCDEFGHI + J at distance 11

Current distance matrix: 2 clusters → 1 clusters

CURRENT DISTANCE MATRIX

–	ABCDEFGHI	J
ABCDEFGHI	0	11
J	11	0

Teal cells mark the smallest non-zero distance.

POINT JOINS A CLUSTER

DISTANCE CALCULATION

d(ABCDEFGHI,J) = 11
Closest pair: I and J
|29 – 40| = 11

RESULTING CLUSTERS

ABCDEFGHIJ

All 10 points are now in one cluster

All nine merges in the correct order

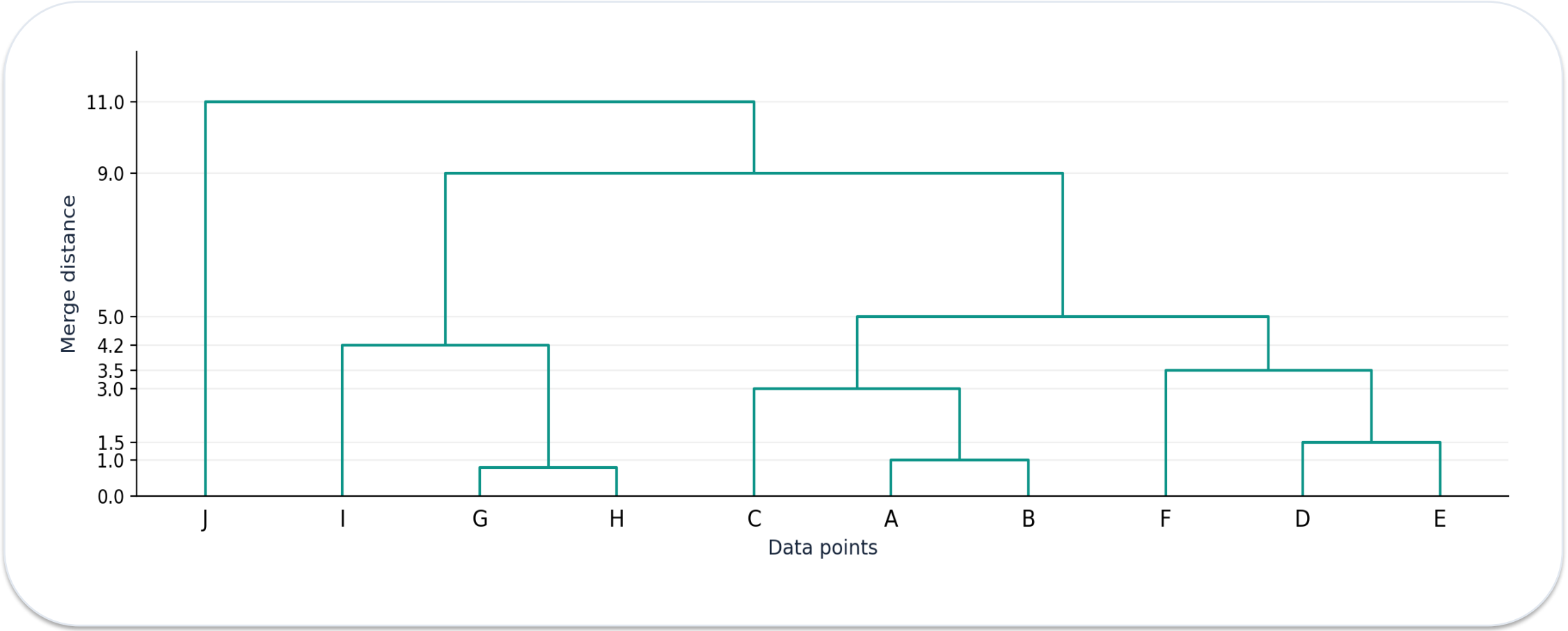
A point can join a cluster; two existing clusters can also merge.

Step	Clusters merged	Distance	Merge type
1	G + H	**0.8**	point + point
2	A + B	**1**	point + point
3	D + E	**1.5**	point + point
4	AB + C	**3**	point + cluster
5	DE + F	**3.5**	point + cluster
6	GH + I	**4.2**	point + cluster
7	ABC + DEF	**5**	cluster + cluster
8	ABCDEF + GHI	**9**	cluster + cluster
9	ABCDEFGHI + J	**11**	point + cluster

Green rows show a single data point joining an already-formed cluster.

The same nine merges form one tree

Height on the dendrogram equals the Single-Linkage merge distance.



One dendrogram can produce different cluster counts

Cut horizontally between merge heights; different heights give different groups.

Cut = 4.5	4 clusters	ABC | DEF | GHI | J
Cut = 6	3 clusters	ABCDEF | GHI | J
Cut = 10	2 clusters	ABCDEFGHI | J